COURSE OUTLINE[1]

Course Title:	FULL STACK DEVELOPMENT
Year of Designing	*2025*
Course Code:	*DAT501*
Credits:	
Faculty Name(s):	**Keerthi Kumar B R**
School:	**CHANAKYA UNIVERSITY**
Email IDs of Coordinators and Instructors:	**keerthi281293@gmail.com**
Level:	*· 400: UG. Advanced. With prerequisites. (PG and PhD students get credit upon School recommendation)*

[1] *Please delete all footnotes and text in grey fonts when submitting the course outline. The grey text is only for your easy reference.*

Course Short Description: (300-500 words)	The course starts by introducing the core concepts of Full Stack Development, covering both the client-side (frontend) and server-side (backend) components. Students will first learn Python, a powerful and easy-to-learn programming language, followed by Django, a high-level web framework that simplifies backend development through its Model-View - Template(MVT) architecture. Alongside Django, students will use PostgreSQL as the backend database to manage and query data efficiently. On the frontend side, the course focuses on building visually appealing and responsive user interfaces using HTML, CSS, JavaScript, and most importantly, React — a popular JavaScript library for building component-based user interfaces. React concepts such as props, state, events, forms, hooks, and conditional rendering will be explored in-depth. A major focus of the course is API development and integration. Students will build RESTful APIs using Django REST Framework and connect them with the frontend using fetch() or axios. This gives students real-time experience in building and consuming APIs – a critical skill in the modern development world. To simulate industry practices, the course includes project-based learning, where students will build a CRUD-based Student Management System. This project will cover real-world features such as data creation, retrieval, update, deletion, API consumption. Students will also be introduced to essential development tools like Git & GitHub (for version control), Postman (for API testing) giving them end-to-end exposure from backend to front end development. This course is ideal for students who aspire to become full stack developers, backend engineers, or front end specialists. By the end of the course, students will have the ability to build test fully functional web applications independently.
Pr-requisites:	**Python, JavaScript, HTML, CSS, DBMS concepts and working knowledge of using SQL**
Pedagogy:	**The course uses a hands-on, project-based learning approach beyond traditional lectures. Students will build real-time projects in every session, practice live coding, and participate in interactive Q&A and code reviews. Tools like GitHub, Postman are used to simulate real-world development. Weekly tasks, peer learning, and one end-to-end full stack project help reinforce concepts practically.**
Expectation from Students:	**Basic Knowledge on OOPS concepts, Data Types(Array, String, Integer, Float), Designing of web pages using HTML and JavaScript and some DBMS concepts would help student to understand Quickly.** **As this course is an hand on course students are expected to focus on developing code, logical thinking and focusing on designing user experienced based case studies.**

Type of Assessment	Marks (total 100)	Description	Schedule (approximate dates)
Project	*10*	*Complete end to end CRUD operation using react and Django*	*Nov 1st 2025*
Assessment on Python , SDLC and Data Structures in Python	*10*	*Includes Objectives, Problem Solving and Theory*	*Aug 9th or Aug 16th 2025*
Assessment to write Queries in Django, Git Commands	*10*	*Includes Objectives, Problem Solving and Theory*	*Aug 30th 2025*
Mid Semester Examination(SDLC, Python, PostgreSQL, Git)	*20*	*Includes Objectives, Problem Solving and Theory*	*Sep 13th 2025*
Assessment on HTML , CSS, JavaScript and React	*10*	*Includes Objectives, Designing and Theory*	*Sep 27th 2025*
Exam on Full Stack(Backend and Front End Technologies)	*40*	*Includes Objectives, Problem Solving and Theory*	*Oct 18th 2025*
Textbooks:	*Will provide some handwritten e-PDF notes on Python, Django and DBMS*		
Reference Materials:	*Python: https://docs.python.org/3/c-api/index.html* *Django: https://docs.djangoproject.com/en/5.2/* *PostgreSQL: https://www.postgresql.org/docs/* *HTML and CSS: https://www.w3schools.com/html/* *React: https://react.dev/learn*		
Digital Resources:	*(Details of video resources with links, if required. Please also add details of MOOC courses that could be used as supplementary study material for this course, for preparatory, additional, or advanced study)*		
Additional Information:	*(Programs in which this course is offered, whether it is a core course or an elective, if this is known, and any other information, expectation from students, including attendance requirements if any etc)*		

Session Plan[2]		
Session No.	**Topic Title with details** **(including assessment days and assessment components)**	**Work plan** **Readings, Cases, Audiovisuals, etc. Activities in class, including lectures, tutorials. Resources and activities for student-work outside class (approx. two hours for every hour of class)**
1	*Introduction & Fundamentals* (3 hrs)	Overview of Full Stack Development Understanding Frontend, Backend, and Full Stack roles Software Development Lifecycle (SDLC) Phases: Requirement, Design, Development, Testing, Deployment
2	*Python Basics (3 hrs)*	Syntax, Variables, Data Types, Rules Control Structures (if, for, while) Functions and Modules
3	*Python Intermediate (3 hrs)*	Object-Oriented Programming Exception Handling File Handling, Debug Virtual Environment and Pip
4	*Version Control (3 hrs)*	Git installation, cloning Git Commands(pull, push, add, commit, diff, status) How to maintain a versions in git
5	*Django Basics (3 hrs)*	Introduction to Django Framework Django Installation & Project Structure URL Routing, Views, Templates, Static Files
6	*Django Intermediate (3 hrs)*	Models, ORM, Forms. Django REST Framework (API Basics) JSON
7	*PostgreSQL and Basic Queries(3 hrs)*	Installing PostgreSQL and pgAdmin Basic SQL (SELECT, INSERT, UPDATE, DELETE) Joins and Relationships

[2] *One-credit course will have 15 hours of teaching, excluding assessment. Please add rows as required.*

8	*Backend Project using Django and PostgreSQL as Database and create Restful API's (3 hrs)*	**Project will involve:** Building a functional backend using Django, integrated with PostgreSQL as the database, focusing on creating Restful API's to perform CRUD operations. Key Points: Connecting PostgreSQL with Django Creating ORM based Queries to perform CRUD Operation(Example on Student or Employee Database)
9	*HTML & CSS & JavaScript Basics (3 hrs)*	HTML Structure and Tags Forms and Tables CSS Basics: Syntax Call API using JavaScript a basic project
10	*React Basics (3 hrs)*	React Setup. Components, Props, State
11	*React Intermediate (3 hrs)*	Event Handling Fetch API and Integrating with Django Backend React Router Basics, JavaScript
12	*Frontend Project to consume Django Restful API's using React and make a complete working CRUD Project (3 hrs)*	This project will focus on developing a responsive frontend using React to consume Django RESTful APIs, enabling full CRUD functionality with real-time interaction and dynamic UI updates.